

Between-Blocks Team Assignment 1— Solutions

XMBA 200S - Fall 2026

OVERVIEW

Copper Kettle Coffee hires SentinelQA to measure store quality. In its Q3 Executive Briefing, SentinelQA reports that the chain's quality index fell from 78.1 in Q1 to 74.2 in Q3, identifies fourteen stores as statistically out of control, and recommends performance-improvement plans, bonus withholdings, and an expanded remediation contract. The purpose of the assignment is to audit those claims using the raw visit data, the store-level data, and SentinelQA's own methods documentation.

Part A

A1. Data audit

The audit identifies several problems that matter for interpreting the briefing. The table below reports the finding, its exact count, the briefing result it affects, and the clarification we would request from SentinelQA.

Finding	Exact count	Briefing result affected	Question for SentinelQA
Undocumented adj_score variable	2,071 non-professional rows	Exhibit 2: Store Excellence Rankings	What is the exact formula for adj_score, why is the adjustment appropriate, and when was the rule established?
Duplicate visit_id values	120 duplicated visit IDs	Exhibit 1 Q2 and Exhibit 4 year-to-date visit count	Why were these records counted twice, and were published statistics computed before or after deduplication?
Undocumented shopper_type = qa_internal	25 rows	Exhibit 2, especially S029	What is qa_internal, and why are these observations included in store-performance rankings?
Suspicious shopper V9999	25 visits; all at S029; all overall_score = 100; all wait_time_min = 1.0	Exhibit 2: S029 is named Store of the Quarter	What was the purpose of these visits, and should internal QA observations be treated as customer-quality visits?

Visits during renovation closure	3 visits at S055	Any statistic using those visits	How could mystery-shop visits occur while the store was closed, and should these rows be excluded?
Different score definitions across exhibits	Exhibit 2 uses adj_score; Exhibits 1 and 3 use overall_score	Cross-exhibit comparisons	Why are different definitions of quality used across exhibits, and which measure should management rely on?

The most consequential finding concerns S029, which SentinelQA labels Store of the Quarter. All 25 qa_internal observations come from shopper V9999, all are assigned to S029, and all have an overall_score of 100. These observations therefore mechanically make S029 appear exceptional. Without them, S029 is not an unusually high-performing store.

The duplicate records also affect reported quantities. The 120 duplicated visit IDs come from an April East-region batch. Keeping them produces the Q2 sample size reported in the briefing. After deduplication, Q2 falls from 1,697 to 1,577 visits, and the Q2 average changes slightly.

A2. What is adj_score?

Comparing adj_score with overall_score reveals the following rule:

adj_score = overall_score for professional shoppers

adj_score = overall_score + 2 for non-professional shoppers

Although the arithmetic can be inferred from the data, the adjustment is not auditable because SentinelQA does not document why two points are added, how that number was estimated, or whether the rule was fixed before the results were observed. A client should require the calibration formula, its justification, a change log, and the ability to reproduce the rankings.

A3. Incentives and measurement

SentinelQA computes the quality measures and determines which stores are classified as underperforming, while it also sells remediation services to flagged stores. This creates a potential conflict of interest because more flags can generate more remediation engagements and therefore more revenue for SentinelQA.

Part B

Prompt output. After dropping duplicate visit IDs, the requested tables are as follows.

Table 1. Mean overall_score by month and shopper program

Month	Professional: n	Professional: mean	App volunteer: n	App volunteer: mean
Jan	480	78.14	-	-
Feb	480	78.17	-	-
Mar	480	77.98	-	-
Apr	480	77.82	-	-
May	480	78.13	-	-
Jun	-	-	617	74.81
Jul	-	-	632	74.34
Aug	-	-	637	74.20
Sep (through Sep 7)	-	-	160	73.65

The professional-shopper mean is essentially flat from January through May. The largest break occurs between May and June, exactly when the shopper program and scoring rubric change.

Table 2. Daypart mix by shopper program and Q3 score by daypart

Daypart	Professional n	Professional share	App volunteer n	App volunteer share	Q3 n	Q3 mean overall_score
Morning	800	33.33%	194	9.48%	143	76.82
Midday	800	33.33%	630	30.79%	431	76.72
Evening	800	33.33%	1222	59.73%	855	72.49
Total	2400	100%	2046	100%	1429	74.20

B1. Is chain quality actually falling?

SentinelQA reports that the Chain Quality Index fell from 78.1 in Q1 to 74.2 in Q3, a decline of 3.9 points. To assess whether this reflects a true deterioration in store quality, we separate the decline into two components.

1. Effect of the change in daypart mix. Under the professional shopper program, visits were approximately balanced across morning, midday, and evening. Under the Community Shopper program, visits are much more concentrated in the evening, when scores are lower. The Q3 average scores by daypart are approximately:

- Morning: 76.8
- Midday: 76.7
- Evening: 72.5

If Q3 had been sampled using the old balanced schedule, with one-third of visits in each daypart, the reweighted Q3 mean would be: $(76.8 + 76.7 + 72.5) / 3 = 75.3$

The actual Q3 mean is 74.2. Therefore, the shift toward evening visits lowers the measured Q3 average by approximately: $75.3 - 74.2 = 1.1$

Thus, about 1.1 points of the reported 3.9-point decline are explained by the change in when stores are visited, rather than by a decline in store quality.

2. What explains the remaining decline?

After accounting for the daypart mix, approximately: $3.9 - 1.1 = 2.8$ points remain. To understand this remaining difference, we examine the monthly averages separately by shopper program:

Month	Shopper program	Mean overall score
January	Professional	78.14
February	Professional	78.17
March	Professional	77.98
April	Professional	77.82
May	Professional	78.13
June	App volunteer	74.81
July	App volunteer	74.34
August	App volunteer	74.20
September*	App volunteer	73.65

The professional-shopper averages are essentially flat from January through May, at about 78. The large drop occurs between May and June, exactly when SentinelQA changes from professional shoppers to Community Shoppers and introduces the new scoring rubric. Therefore, the comparison between Q1 and Q3 is not an apples-to-apples comparison of underlying store quality.

In conclusion, the data do not support interpreting the full 3.9-point decline as a true deterioration in chain quality.

B2. Who is in the sample after June 1?

Places. The Community Shopper program heavily over-samples university-area stores: 24 university stores make up 20% of the chain but account for 781 of 1,429 Q3 visits, or 54.7%. Their Q3 average is 73.92

versus 74.53 for non-university stores, so this place-composition shift pushes the visit-weighted chain average downward.

Times. The professional program is perfectly balanced across dayparts, while 59.7% of app-volunteer visits occur in the evening. Q3 evening visits average only 72.49, compared with 76.82 in the morning and 76.72 at midday. This timing shift therefore lowers the measured chain average.

People and measurement. The pre-June observations come from trained professional shoppers on a fixed schedule, whereas the post-June observations come from self-selected app users who choose when and where to shop. At the same time, SentinelQA changes to rubric v2. The resulting difference combines shopper selection and a change in the measurement instrument with any possible change in true store quality.

Part C

Q3 chain-wide mean (app volunteers, July 1-September 7)

Statistic	Value
Mean ($\bar{x}$)	74.2005
Sample standard deviation (s)	7.2001
n	1,429
Standard error	$7.2001 / \sqrt{1,429} = 0.1905$
95% margin of error	$1.96 \times 0.1905 = 0.3733$
95% confidence interval	[73.827, 74.574]

$$CI = \bar{x} \pm 1.96 \times (s / \sqrt{n}) = 74.2005 \pm 1.96 \times (7.2001 / \sqrt{1429}) = 74.2005 \pm 0.3733$$

95% CI = [73.827, 74.574]

The briefing misinterprets the interval when it states: “Statistically, 95% of customer experiences now fall between 73.8 and 74.6.” This is incorrect because a confidence interval is an interval for the population mean, not an interval containing 95% of individual customer experiences. A correct statement is: “Using this sampling procedure, the estimated Q3 mean is 74.2, with a 95% confidence interval of approximately 73.8 to 74.6.”

S048 Q3 mean

Statistic	Value
Q3 scores	67.9, 77.2, 59.6, 72.0
Mean ($\bar{x}$)	69.175
Sample standard deviation (s)	7.4316
n	4
Degrees of freedom	3
t critical value, 95%	3.182
Standard error	$7.4316 / \sqrt{4} = 3.7158$
95% confidence interval	[57.35, 81.00]

$$CI = \bar{x} \pm t \times (s / \sqrt{n}) = 69.175 \pm 3.182 \times (7.4316 / \sqrt{4}) = 69.175 \pm 11.826$$

95% CI = [57.35, 81.00]

The interval is very wide for two reasons. First, with only four observations the standard error is large. Second, because the standard deviation is estimated from such a small sample, the t critical value is much larger than 1.96, adding further uncertainty. The interval contains both the Q3 chain mean of 74.2 and values near the earlier chain level of 78. Therefore, these four observations do not provide precise evidence that S048 is truly a low-quality store. They also do not prove that S048 is healthy; rather, the data are too imprecise to support a strong store-level conclusion or a managerial penalty.

C3. Margin of error and required sample size

Using the class approximation $n = 4\sigma^2 / \text{MoE}^2$, with $\sigma = 6.5$ and a target margin of error of ± 2 points:

$$n = 4(6.5^2) / 2^2 = 42.25$$

Because sample size must be an integer and we need to meet the target precision, we round up to 43 visits per store per quarter.

$$43 \text{ visits} \times 120 \text{ stores} \times \$40 = \$206,400 \text{ per quarter}$$

$$\$206,400 \times 4 = \$825,600 \text{ per year}$$

Thus, achieving approximately ± 2 -point precision for every store would cost about \$206,400 per quarter, or \$825,600 per year, at \$40 per professional visit. That is substantially above Copper Kettle's current \$480,000 annual contract and also above SentinelQA's proposed \$700,000 expanded program.

Part D

Prompt 1 output. SentinelQA uses the lower control limit:

$$\text{LCL} = 74.2 - 2 \times (6.5 / \sqrt{16}) = 70.95$$

Using Q3 app-volunteer visits (July 1-September 7), the same fourteen stores are flagged:

Store	Q3 mean	Actual Q3 n
S083	65.95	33
S017	67.24	37
S066	68.50	4
S049	69.02	6
S048	69.18	4
S044	69.63	8
S001	69.87	6
S071	70.01	8
S057	70.03	3
S041	70.40	11
S120	70.40	4
S086	70.59	7
S054	70.66	5
S092	70.83	3

The key problem is immediately visible: the fourteen stores do not each have 16 visits. Several have only 3-8 visits, so using the same standard error for every store understates uncertainty for low-volume stores.

Prompt 2 output. The corrected store-specific lower control limit is:

$$LCL_store = 74.2 - 2 \times (6.5 / \sqrt{n_store})$$

With this correction, only three stores fall below their own two-standard-error limit:

Store	Q3 mean	n	Own-n lower limit
S083	65.95	33	71.94
S017	67.24	37	72.06
S015	71.63	29	71.79

For a healthy store with true mean 74.2 and $n = 7$:

$$SE = 6.5 / \sqrt{7} = 2.46$$

$$z = (70.95 - 74.2) / 2.46 = -1.32$$

$$P(\text{flagged by chance}) = \Phi(-1.32) = 0.093 \approx 9.3\%$$

Even if the store is completely healthy, SentinelQA's rule will incorrectly classify it as "out of control" about 9 times out of 100 quarters, simply because its sample is small.

D1. Store-level verdicts

Store	Q3 mean	n	Verdict	Evidence	Recommended action
S083	65.9	33	Real problem	H1 mean 80.3; very low Q3 mean even with large n.	Act now; investigate and begin corrective action.
S017	67.2	37	Real problem	H1 mean 78.7; large and precisely measured Q3 decline.	Act now; investigate and consider withholding bonus subject to D3.
S066	68.5	4	False alarm	Only 4 Q3 visits; own-n uncertainty is very large.	Collect more data; no PIP or bonus action now.
S049	69.0	6	False alarm	Only 6 visits; H1 mean 75.9.	Collect more data before acting.
S048	69.2	4	False alarm	Only 4 visits; its 95% CI is extremely wide.	No adverse action; collect more visits.
S044	69.6	8	Artifact + luck	All Q3 visits were in the evening; H1 mean 74.9.	Resample across dayparts before acting.
S001	69.9	6	False alarm	Only 6 visits; H1 mean 77.0.	Collect more data.
S071	70.0	8	Artifact + luck	All Q3 visits were in the evening; H1 mean 77.3.	Resample across dayparts before acting.
S057	70.0	3	False alarm	Only 3 Q3 visits.	Do not act on this sample.
S041	70.4	11	False alarm	Does not survive own-n control limit; H1 mean 78.8.	Monitor; no PIP now.
S120	70.4	4	False alarm	Only 4 Q3 visits.	Collect more data.
S086	70.6	7	False alarm	Does not survive own-n limit.	Monitor; no adverse action.
S054	70.7	5	False alarm	Only 5 Q3 visits.	Collect more data.
S092	70.8	3	Artifact / false alarm	Only 3 visits, all in the evening.	Resample across dayparts; no action now.
S015	71.6	29	Corrected-rule flag	Falls just below its own two-SE limit; not flagged by vendor.	Investigate, but treat cautiously because a 2-SE rule creates expected false positives.

D2. Why the vendor's rule creates false alarms

For a store with $n = 7$, the correct standard error is: $SE = 6.5 / \sqrt{7} = 2.46$

The vendor's fixed lower limit is 70.95, which is 3.25 points below 74.2. In units of the store's true standard error:

$$(74.2 - 70.95) / 2.46 = 1.32 \text{ standard errors}$$

Therefore, a completely healthy $n = 7$ store has about a 9.3% chance of being flagged in any quarter:

$$P(Z < -1.32) = 0.093$$

With 96 low-volume stores, the expected number of chance flags is approximately:

$$96 \times 0.093 = 8.9, \text{ or about 9 stores per quarter}$$

Under the vendor's rule, a healthy chain is expected to generate roughly nine low-volume flags by chance alone each quarter.

D3. Rule for next quarter and one-sided bounds

For next quarter, we would set the control rule before seeing the data. We would use each store's actual number of visits in the standard error, require at least 43 visits per store per quarter as calculated in C3, and use a more conservative 3-standard-error lower control limit:

$$LCL = \text{chain mean} - 3 \times (6.5 / \sqrt{n})$$

This is preferable to SentinelQA's current rule because the current rule assumes $n = 16$ for every store and therefore produces too many false alarms among low-volume stores.

Before withholding a manager's bonus, however, we would use a stricter evidentiary standard. Specifically, we compute the one-sided 95% upper bound for the store's true quality:

$$\text{Upper bound} = \bar{x} + 1.645 \times (6.5 / \sqrt{n})$$

The idea is to give the store the most favorable interpretation consistent with the data. If even this upper bound is still clearly low, then the adverse action is supported.

For S017:

$$SE = 6.5 / \sqrt{37} \approx 1.07$$

$$\text{Upper bound} = 67.2 + 1.645 \times 1.07 \approx 69.0$$

Even the most favorable plausible value for S017's true quality is only about 69.0, well below the chain mean of 74.2. Therefore, the evidence is strong enough for the bonus withholding to hold up.

For S048:

$$SE = 6.5 / \sqrt{4} = 3.25$$

$$\text{Upper bound} = 69.2 + 1.645 \times 3.25 \approx 74.5$$

For S048, the true quality could plausibly be as high as about 74.5, which is consistent with a healthy store. Therefore, withholding the bonus is not supported by the available data.

D4. A failure the quarterly-average chart can miss

A control chart based only on the quarter average can miss a store that is deteriorating steadily within the quarter. A high score early in the quarter can offset much lower scores later and leave the quarterly mean above the threshold.

S061 illustrates this problem. Its Community Shopper monthly averages decline sharply:

Month	Mean overall score	n
June	80.1	15
July	75.1	24
August	73.0	15
September through Sept. 7	64.0	5

S061's Q3 average is about 73.1 with $n = 44$, so it is not flagged by either the vendor's fixed threshold or the corrected two-SE threshold. Yet the monthly path shows a clear within-quarter decline. This is exactly the type of failure a threshold on the quarter average can miss.

Part E

E1. Memo (next page)

E.2

Before attributing S017's decline to the new manager, we would need to know whether the store was already declining before July 1, what else changed at the store around the same time, and how comparable stores without a management change performed over the same period. The data show that S017's measured quality is much lower in Q3, but the timing alone does not establish that the manager caused the decline. Without a credible comparison or counterfactual showing what would have happened at S017 without the management change, this dataset cannot identify the manager's causal effect.

To: CEO, Copper Kettle Coffee Co.

From: Group XX

Subject: Memo Q3 SentinelQA

Recommendation. Do not treat the reported 78.1-to-74.2 decline as evidence that chain quality fell by 3.9 points, and do not place all fourteen managers on PIPs or withhold all fourteen bonuses. The evidence is strong enough to act now at S017 and S083; the remaining flagged stores require additional, better-balanced data. We would not approve SentinelQA's \$700,000 Expanded Coverage & Remediation proposal under the current methodology.

What the decline means. The Q1 and Q3 averages are not directly comparable because SentinelQA changed both the shopper program and scoring rubric on June 1. About 1.1 points of the 3.9-point decline are explained by the shift toward lower-scoring evening visits, while the remaining drop appears as a discrete break at the program change rather than a gradual decline in store quality.

Exhibit 1. Store-level action summary

Store/group	Q3 mean	n	Assessment	Recommended action
S083	65.9	33	Real deterioration; large-n result	Act now; investigate and withhold bonus subject to normal review
S017	67.2	37	Real deterioration; one-sided 95% upper bound ≈ 69.0	Act now; evidence survives a manager-favorable bound
S048	69.2	4	Too little information; one-sided upper bound ≈ 74.5	No bonus withholding; collect more visits
Other 11 vendor flags	69.0-70.8	3-11	Mostly small-n false alarms and/or daypart artifacts	No immediate PIP; resample with balanced coverage
S015	71.6	29	Flagged only under corrected own-n 2-SE rule	Investigate, but do not punish on a screening flag alone

Bonus decision. SentinelQA applies the same cutoff to every store even though many have very few visits, so healthy low-volume stores are flagged too often. With only 7 visits, a healthy store has about a 9% chance of being flagged by mistake, implying roughly 9 false alarms among the 96 low-volume stores. We therefore treat flags as a reason to investigate, not to punish automatically. S017 still looks clearly weak even under a favorable statistical bound; S048 does not.

Contract recommendation. Any renewal should condition payment on a transparent and pre-specified measurement protocol. At minimum, require:

- a documented scoring rubric and calibration formula, with a dated change log;
- store-level control limits based on each store's actual n, fixed before the quarter begins;
- a minimum of 43 visits per store per quarter before performance decisions are made;
- balanced coverage across dayparts and a clear separation of internal QA observations from customer-quality visits; and
- monitoring of within-quarter trends, not only quarterly averages, so deteriorating stores such as S061 are not missed.